

The Guardian: A Real-Time Browser-Based Hate Speech Detection Framework Using Transformer Models

Randinu Fernando

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
IT22366054@my.sliit.lk

Gimhani Herath

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
it22361486@my.sliit.lk

Hiruni De Alwis

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
IT22317926@my.sliit.lk

Vishwa Rathnayaka

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
vjr200124@gmail.com

Kavinga Abeywardena

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
kavinga.y@sliit.lk

Deemantha Siriwardana

Dept. of Computer Systems Engineering
Sri Lanka Institute of
Information Technology
Malabe, Sri Lanka
deemantha.s@sliit.lk

Abstract—The rapid growth of social media platforms has transformed digital communication by enabling real-time interaction among millions of users worldwide. However, this growth has also increased the spread of harmful online content, particularly hate speech and offensive language. Such content can negatively impact individuals by promoting harassment, discrimination, psychological distress, and unsafe online environments. Traditional moderation systems implemented by social media platforms often operate reactively, meaning harmful content may remain visible before being removed. This creates a significant gap in real-time user protection.

This research proposes a real-time hate speech detection framework integrated into The Guardian; a browser-based cybersecurity solution designed to protect users from harmful online interactions. The proposed system automatically detects textual content from social media platforms, analyzes the extracted text using a transformer-based machine learning model, and classifies content into three categories: hate speech, offensive content, and neutral content.

The machine learning component uses a fine-tuned DistilBERT model trained using multiple datasets including the Davidson dataset, HateXplain dataset, and Jigsaw Toxic Comment dataset. During development, several challenges such as dataset imbalance, label inconsistency, and prediction bias were identified and addressed through dataset balancing, preprocessing improvements, and confidence calibration.

The final system provides real-time browser-side protection by highlighting harmful content using visual indicators, generating confidence scores, and alerting users before interaction occurs. The research demonstrates how natural language processing techniques can be integrated into cybersecurity systems to improve user safety in modern social media environments.

Index Terms—Hate Speech Detection, Cybersecurity, DistilBERT, Social Media Security, Browser Extension, Offensive Language Detection, Natural Language Processing

I. INTRODUCTION

Social media platforms such as Facebook, Instagram, X, and TikTok have become central to modern communication, enabling users to share information, express opinions, and engage in global discussions. However, the rapid growth of these platforms has also led to the widespread dissemination of harmful online content, particularly hate speech and offensive language. Such content can negatively impact individuals and communities by promoting discrimination, harassment, and psychological harm.

Hate speech detection has become an important research area within natural language processing due to the increasing prevalence of toxic online behavior. Early studies have shown that distinguishing between hate speech and offensive language is a complex task because both categories often share similar vocabulary and linguistic structures [1]. In addition, existing datasets frequently contain class imbalances and labeling inconsistencies, which further complicate classification performance [6], [7].

Traditional approaches to hate speech detection relied on machine learning algorithms such as Support Vector Machines and Naïve Bayes classifiers. While these methods provided initial progress, they required extensive feature engineering and struggled to capture contextual meaning in text [6]. More recent approaches have shifted toward deep learning models, which automatically learn language patterns from data and improve classification accuracy [8].

The introduction of transformer-based models significantly advanced natural language processing tasks. Models such as BERT: Pre-training of Deep Bidirectional Transformers for

Language Understanding demonstrated the ability to capture bidirectional context in language, resulting in improved performance across multiple NLP tasks [3]. Subsequent work introduced more efficient models such as DistilBERT, which reduce computational complexity while maintaining strong performance, making them suitable for real-time applications [4].

Despite these advancements, most existing hate speech detection systems are designed for offline analysis and platform-level moderation. In many cases, harmful content is detected only after it has already been posted and viewed by users. This creates a reactive moderation model that does not prevent initial exposure to harmful content. Additionally, many systems lack transparency, providing classification outputs without indicating prediction confidence.

From a cybersecurity perspective, harmful online communication represents a human-layer threat. Unlike traditional cyber-attacks that target systems and networks, hate speech directly affects users by influencing emotions, behavior, and online safety. Therefore, there is a need for solutions that provide proactive protection at the user level rather than relying solely on centralized moderation systems.

To address these challenges, this research proposes a real-time hate speech detection framework integrated into a browser extension. The system automatically detects visible social media content, classifies text using a transformer-based model, and provides immediate feedback to users through visual indicators and confidence scores. By combining natural language processing with practical deployment mechanisms, the proposed system aims to enhance user awareness and improve protection against harmful online interactions.

The key objectives of this research are:

- Develop a real-time hate speech detection system.
- Distinguish between hate speech, offensive language, and neutral content.
- Provide confidence-based classification transparency.
- Implement browser-based protection mechanisms.
- Improve user safety in social media environments.

II. LITERATURE REVIEW

Hate speech detection has become an important research area within natural language processing due to the increasing volume of harmful content on social media platforms. Early research primarily focused on traditional machine learning approaches, which relied on manually engineered features such as Bag-of-Words and TF-IDF representations. Methods such as Support Vector Machines and Logistic Regression were widely used in initial studies, but these approaches struggled to capture contextual meaning and semantic relationships in text [6].

One of the key challenges identified in early research is the difficulty of distinguishing between hate speech and offensive language. Studies have shown that these categories often overlap in vocabulary and structure, making classification tasks more complex [1]. Similarly, research by Zeerak Waseem and Dirk Hovy highlighted that hate speech is not always explicitly

expressed and may depend heavily on context and target identification, which further complicates automated detection [2].

With the advancement of deep learning, researchers began applying neural network architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to improve classification performance. These models demonstrated better ability to learn patterns directly from data without requiring extensive feature engineering [8]. Hierarchical models and hybrid approaches were also explored to capture both word-level and sentence-level information, improving detection accuracy in complex linguistic scenarios [11].

A major breakthrough in natural language processing came with the introduction of transformer-based architectures. The transformer model proposed in Attention Is All You Need introduced self-attention mechanisms that significantly improved the ability to capture long-range dependencies in text [15]. Building on this foundation, models such as BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding enabled bidirectional contextual learning, which greatly enhanced performance in text classification tasks [3]. Subsequent work introduced more efficient variants such as DistilBERT, which reduce model size while maintaining strong performance, making them suitable for real-time applications [4].

Transformer-based models have since been widely applied to hate speech detection tasks. Studies have shown that fine-tuned BERT models outperform traditional machine learning and earlier deep learning approaches in identifying harmful language [10]. However, despite these improvements, model performance is still highly dependent on dataset quality and diversity.

Dataset limitations remain a significant challenge in hate speech research. The Davidson dataset is one of the most commonly used datasets in this domain, but it contains a strong imbalance between offensive and hate speech classes, which can bias model predictions [1]. To address such limitations, newer datasets such as HateXplain have been introduced, providing more detailed annotations and improved representation of hate speech examples [5]. Large-scale datasets collected through crowdsourcing have also contributed to improving model generalization by providing more diverse language patterns [12].

Another important issue in hate speech detection is the lack of consistent definitions across datasets. Research has shown that labeling inconsistencies can negatively affect model performance, particularly when combining multiple datasets [13]. This highlights the importance of preprocessing and label standardization when developing robust classification systems.

Despite the significant progress in model development, most existing studies focus primarily on improving classification accuracy in offline environments. There is limited research on deploying hate speech detection systems in real-time user environments. Many existing solutions operate at the platform level, where content moderation occurs after harmful content

has already been posted and viewed.

This research addresses this gap by combining transformer-based classification with browser-based deployment, enabling real-time detection and user-side protection. By integrating machine learning with practical cybersecurity mechanisms, the proposed system extends beyond traditional classification approaches and focuses on improving user safety during live social media interactions.

III. METHODOLOGY

This research follows a multi-stage methodology that combines dataset preparation, transformer-based model training, backend integration, browser extension deployment, and real-time user alert generation. The overall objective of the methodology was to develop a practical hate speech detection system capable of identifying harmful social media content in real time while maintaining high classification efficiency.

The development process began with dataset selection and preprocessing, followed by iterative model training and evaluation. Once acceptable classification performance was achieved, the trained model was integrated into a backend API and connected to a browser extension for real-time deployment.

Unlike many prior studies that focus solely on offline classification accuracy, this methodology emphasizes practical deployment challenges such as latency, prediction transparency, browser compatibility, and user safety.

A. Dataset Selection and Initial Challenges

The initial training phase relied on the Davidson dataset due to its widespread use in hate speech classification research. The dataset contains approximately 24,783 labeled social media text samples categorized into hate speech, offensive language, and neutral content.

The Davidson dataset was initially selected because of its popularity in academic literature and its relevance to short-form social media text classification. However, early training experiments revealed significant limitations.

During initial testing, the model demonstrated strong performance in identifying offensive language but frequently failed to detect explicit hate speech. This occurred because the dataset contains a significantly larger number of offensive samples compared to hate speech samples. As a result, the model developed a classification bias toward offensive predictions.

For example, highly aggressive statements such as “You people should be eliminated from society” were incorrectly classified as neutral or offensive content during early experiments. These results highlighted the limitations of relying on a single dataset for real-world deployment.

B. Multi-Dataset Integration

To address the limitations identified during initial experimentation, additional datasets were introduced to improve classification diversity and reduce prediction bias.

The HateXplain dataset was integrated to provide more explicit hate speech examples. This dataset contains carefully annotated samples that include stronger representations of

group-targeted harmful speech. By incorporating HateXplain, the model was exposed to more diverse hate-related language patterns that were underrepresented in the Davidson dataset.

Although HateXplain improved hate speech recognition, the system still faced challenges when processing general toxic comments that did not explicitly target protected groups. To improve broader offensive language recognition, selected samples from the Jigsaw Toxic Comment dataset were incorporated into the training pipeline.

The inclusion of multiple datasets significantly improved the diversity of training samples and helped the model generalize better across different forms of harmful online language.

C. Data Preprocessing and Label Standardization

One of the major challenges in combining multiple datasets was the inconsistency of labeling structures. Each dataset used different classification schemes, making direct integration impossible.

To solve this issue, all datasets underwent extensive preprocessing procedures before model training. Duplicate entries were removed to prevent overfitting caused by repeated examples. Unnecessary symbols, formatting issues, and incomplete text samples were also cleaned.

Most importantly, label standardization was performed to create a unified three-class classification structure:

- Hate
- Offensive
- Neutral

Samples from all datasets were mapped into these categories based on their original labels. This preprocessing stage was critical because improper label alignment could significantly reduce model accuracy during training.

D. Hate Speech Detection Model Training

The machine learning component serves as the analytical core of the proposed system. A transformer-based DistilBERT model was selected due to its balance between computational efficiency and classification performance.

While larger transformer architectures such as BERT and RoBERTa often achieve strong classification accuracy, they require significantly greater computational resources and introduce higher inference latency. Since the proposed system was designed for real-time browser deployment, low-latency inference was a major requirement. DistilBERT was selected because it reduces model size while maintaining strong contextual language understanding capabilities.

The final dataset was tokenized using the DistilBERT tokenizer and divided into training and validation sets. The model was then fine-tuned using PyTorch and Hugging Face Transformers.

Training was conducted in a CPU-based environment, which introduced longer training times compared to GPU environments. Despite these limitations, the training process was successfully completed.

E. Model Optimization and Debugging

Following model training, several deployment issues emerged during real-world testing.

One major issue involved incorrect label mapping configurations. Although the model generated correct probability distributions internally, incorrect mapping of class IDs caused harmful content to be displayed as neutral content. For example, explicit hate speech inputs were incorrectly returned as neutral predictions despite high probability scores for harmful categories. This issue was resolved by correcting the label mapping configuration within the backend system.

A second issue involved confidence calibration. After resolving label mapping errors, the model temporarily began overpredicting hate speech classifications for harmless content. This issue was addressed by refining confidence thresholds and improving probability interpretation logic. These debugging iterations significantly improved final model stability.

F. Real-Time System Deployment Workflow

After achieving stable classification performance, the trained model was integrated into a real-time deployment architecture.

The browser extension automatically scans visible social media content and extracts relevant text from posts and comments. The extracted text is then sent to the FastAPI backend for classification. The backend performs model inference and returns:

- Predicted label
- Confidence score
- Risk classification

The browser extension then visually flags harmful content using color indicators and generates alerts for users. This workflow enables proactive user protection by identifying harmful content before users engage with it.

IV. IMPLEMENTATION

The implementation phase focused on transforming the trained hate speech detection model into a functional real-time cybersecurity tool that can operate within social media environments. While achieving strong classification accuracy was important, the practical deployment of the model required additional components such as browser integration, backend communication, confidence evaluation, and user alert mechanisms.

The final implementation architecture was designed to ensure that harmful content could be detected with minimal delay while users browse social media platforms. The system consists of three major components: the browser extension layer, the backend API layer, and the machine learning inference layer.

A. System Architecture and Technology Stack

The proposed system follows a client-server architecture that enables real-time communication between frontend browser components and backend machine learning services.

At the client side, a browser extension was developed to monitor user interactions within social media platforms. The extension scans visible web content and identifies text segments that require classification. At the server side, a FastAPI backend was developed to receive extracted text from the browser extension and process prediction requests. The backend communicates directly with the trained DistilBERT model. The machine learning layer performs text classification and returns prediction outputs to the browser extension.

The technology stack used in this research was selected based on performance efficiency and ease of deployment. Python was used for backend development due to its strong machine learning ecosystem. FastAPI was selected because it provides lightweight API development with low latency. PyTorch and Hugging Face Transformers were used for model training and inference. JavaScript, HTML, and CSS were used to build the browser extension interface.

This architecture ensured modularity and allowed each system component to be independently improved without affecting the overall workflow.

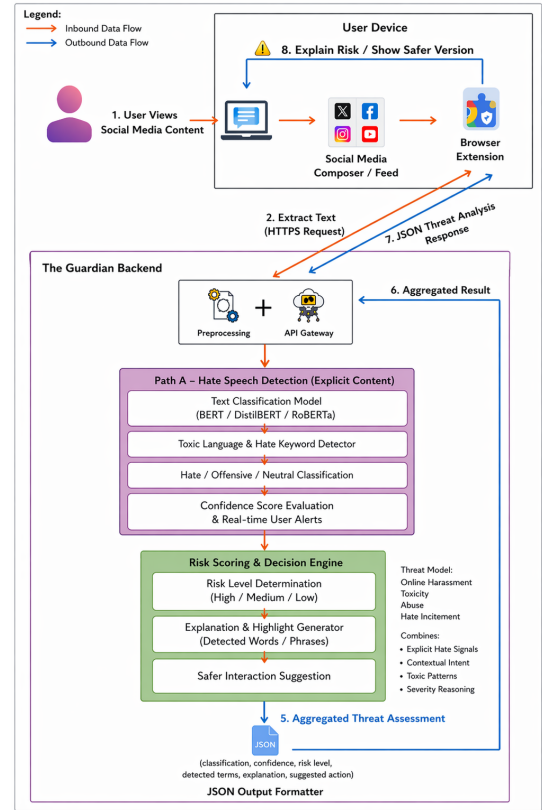


Fig. 1. Overall architecture of The Guardian real-time hate speech detection system.

B. Browser Extension Implementation

The browser extension serves as the primary user-facing component of the proposed system. Unlike traditional moderation systems that operate at the platform level, this extension works directly within the user's browser environment.

The extension continuously scans visible content within social media feeds, posts, and comment sections. Using Document Object Model (DOM) analysis, the extension identifies textual elements that are likely to contain user-generated content. Once text is detected, the extension extracts the content and forwards it to the backend API for analysis.

After receiving classification results, the extension modifies webpage elements in real time by applying visual indicators. Hate speech content is highlighted using red markers, while offensive content is highlighted using orange markers. Neutral content remains unchanged.

In addition to visual highlighting, the extension generates alerts that inform users about detected harmful content and displays the model’s confidence score. This implementation allows users to receive immediate warnings before interacting with harmful content, which supports the proactive security objective of the research.

Real-Time Detection Workflow (The Guardian)

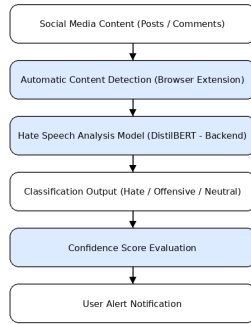


Fig. 2. Real-time detection workflow of The Guardian browser-based system.

C. Backend API Implementation

The backend system acts as the communication layer between the browser extension and the machine learning model. FastAPI was selected because of its lightweight design and ability to handle real-time API requests efficiently. The backend exposes a prediction endpoint that receives extracted text from the browser extension.

Communication between the browser extension and backend system was designed using a structured API contract to reduce integration failures. The API accepts text content through a POST request endpoint and processes JSON-based payloads containing extracted text and contextual metadata such as platform source and timestamps. The response returns detected risks, confidence scores, highlighted text, and processing metadata. This structured contract improved system interoperability between frontend and backend components.

When text is received, the backend performs tokenization using the DistilBERT tokenizer. The processed input is then forwarded to the trained classification model. The model

returns probability distributions for each class, and the backend selects the most probable label as the final classification output.

The backend response includes:

- Predicted classification label
- Confidence score
- Original text input

The API was designed to maintain low response latency since delays would negatively affect user browsing experience.

D. Hate Speech Detection Model Integration

The machine learning model serves as the analytical core of the implementation phase. The final model integrated into the backend was a fine-tuned DistilBERT model trained using combined datasets.

The final model was built using the DistilBERT (distilbert-base-uncased) architecture and trained using supervised fine-tuning techniques through Hugging Face Transformers and PyTorch. The model was trained for three epochs to balance performance improvement and computational efficiency. This configuration was selected because the project required lower inference latency for real-time browser deployment.

Several implementation challenges emerged during model integration. Initially, incorrect label mappings caused explicit hate speech content to be classified as neutral content. Although the internal model probabilities indicated harmful classifications, incorrect label associations produced inaccurate final outputs. This issue became visible when phrases such as “You people should be eliminated from society” were incorrectly returned as neutral predictions. After debugging the model configuration, the label mapping issue was resolved.

A second issue emerged when confidence thresholds caused harmless text to be incorrectly classified as hate speech. For example, normal statements and mild insults occasionally triggered false hate classifications. This issue was resolved by adjusting confidence handling logic and improving probability interpretation. These iterative improvements significantly increased deployment reliability.

E. Confidence Score Evaluation Implementation

One major limitation of many machine learning classification systems is the lack of transparency in prediction outputs. Users often receive classifications without understanding how confident the model is in its decision.

To address this limitation, the proposed system incorporates confidence score evaluation. After classification, the softmax probability values generated by the model are used to determine prediction confidence. The highest probability score is returned alongside the predicted label. For example, if harmful text is detected with a confidence score of 96%, users are informed that the prediction is highly certain. This improves transparency and helps users make informed decisions regarding potentially harmful content.

F. Risk Score Implementation

The system assigns severity levels based on classification output and confidence values. Content classified as hate speech with high confidence is treated as high-risk content. Offensive content receives medium-risk classification, while neutral content is treated as low-risk content. This additional layer helps prioritize warnings and improves the overall usability of the system.

G. Performance and Optimization

Real-time detection systems must maintain strong performance without interrupting normal user browsing behavior. Several optimization strategies were implemented during deployment. DistilBERT was selected instead of larger transformer models to reduce inference latency. FastAPI reduced backend processing delays. The browser extension was optimized to process only visible content instead of scanning entire web pages continuously. These optimizations ensured smooth system performance during real-time testing.

H. Privacy and Security Implementation

Privacy preservation was an important design consideration throughout system development. The browser extension only extracts textual content required for classification and avoids collecting unnecessary personal information.

Privacy preservation was further strengthened through the API design. According to the system contract, user text is processed as temporary data and is not permanently stored in backend systems. Anonymous user identifiers are used instead of personally identifiable information, and all communication is designed to operate through secure HTTPS connections. This aligns with the project's privacy-by-design approach.

The system does not store user browsing history or personal account information. Communication between the browser extension and backend remains limited to temporary classification requests. From a cybersecurity perspective, this design minimizes privacy risks while still providing effective protection against harmful online interactions.

V. EVALUATION AND RESULTS

The evaluation phase was conducted to assess both the machine learning performance of the proposed hate speech detection model and the practical effectiveness of the complete real-time system. Since the proposed framework combines natural language processing with browser-based deployment, evaluating only classification accuracy would not be sufficient. Therefore, this section examines the model's classification capabilities, confidence reliability, real-time browser performance, and the effectiveness of the implemented risk alert mechanism.

A. Evaluation Methodology

The evaluation process was divided into two major stages: model-level evaluation and system-level evaluation.

The first stage focused on evaluating the machine learning model independently. During this phase, the primary objective was to determine whether the DistilBERT model could

accurately classify textual inputs into the three predefined categories: hate speech, offensive content, and neutral content. Multiple test samples were manually entered to observe prediction outputs and identify classification weaknesses.

The final model evaluation was conducted using a structured dataset split consisting of training, validation, and testing datasets. Out of the total 24,783 samples used during the initial training phase, 19,826 samples were allocated for training, 2,478 samples were used for validation, and 2,479 samples were reserved for final testing. The use of a held-out test set ensured that final evaluation results remained unbiased and prevented data leakage during training.

The second stage focused on evaluating the complete deployed system. This involved testing the browser extension's ability to extract social media text, communicate with the FastAPI backend, receive classification outputs, and display visual warnings in real time.

B. Classification Performance

The initial model trained solely on the Davidson dataset produced inconsistent classification results. Although the model performed reasonably well when identifying offensive language, it struggled to correctly detect explicit hate speech content. This issue was primarily caused by the class imbalance present in the Davidson dataset. Since offensive language samples significantly outnumbered hate speech samples, the model developed a bias toward predicting offensive classifications.

During validation testing, the model achieved a validation accuracy of 91.69%, with a validation loss of 0.2784. The macro precision score was recorded as 0.7854, while recall reached 0.7835. The final validation F1-score was 0.7842. These results demonstrated strong classification capability while maintaining balanced performance across multiple classes.

Final testing produced an accuracy of 92.01%, with a test loss of 0.2592. The model achieved a macro precision score of 0.7834, recall of 0.7845, and an F1-score of 0.7831.

After combining all three datasets and performing additional preprocessing, the final model demonstrated significantly improved classification stability. For example:

- The statement "You people should be eliminated from society" was correctly classified as hate speech with a confidence score of 0.96.
- The statement "You are an idiot" was correctly classified as offensive content.
- The statement "Have a nice day everyone" was correctly classified as neutral content.

C. Confidence Validation

Confidence score validation was conducted to determine whether the probability outputs generated by the model accurately reflected prediction certainty. The system uses softmax probabilities to calculate confidence values for each prediction. These confidence scores are displayed directly to users through the browser extension interface.

During testing, highly aggressive and explicit hate speech inputs generally produced high confidence values, indicating strong model certainty. Similarly, clearly neutral statements produced high confidence scores for neutral predictions.

However, early versions of the system experienced confidence calibration issues. After correcting the label mapping problem, the model temporarily began overpredicting hate speech classifications for harmless text. After refining confidence thresholds and correcting label mappings, prediction confidence became significantly more stable and reliable.

D. End-to-End System Performance

The complete system was evaluated in a real browsing environment to determine whether all components functioned correctly as an integrated solution. The browser extension successfully detected visible text from social media content and extracted relevant textual information. The extracted text was transmitted to the FastAPI backend for classification.

Performance testing showed that the model processed evaluation samples at approximately 5.96 samples per second during final testing, with an evaluation runtime of 416.03 seconds. These results indicate that the model maintained acceptable processing speed for real-time deployment requirements.

The full workflow operated as follows:

Social media text extraction → *backend API communication*
→ *model inference* → *confidence generation* → *risk classification* → *user alert notification*

Testing confirmed that this process occurred with minimal noticeable delay, allowing users to continue browsing without major performance interruptions.

E. Behavioural Adaptation Evaluation (Risk Score Evaluation)

During testing, content classified as high-confidence hate speech triggered stronger warnings and red visual indicators. Offensive content triggered moderate warnings and orange indicators. Neutral content produced no warnings. The results showed that the system successfully differentiated between varying levels of harmful content and communicated those risks effectively to users.

F. Summary of Results

The final evaluation demonstrates that the proposed system successfully achieved both technical and practical objectives. From a machine learning perspective, the integration of multiple datasets significantly improved classification accuracy and reduced prediction bias. From a deployment perspective, the browser extension successfully provided real-time harmful content detection without significantly affecting user browsing performance.

VI. DISCUSSION

A. Interpretation of Results

The experimental results demonstrate that transformer-based language models can be effectively applied to real-time hate

TABLE I
PERFORMANCE METRICS OF THE FINAL TRAINED DISTILBERT MODEL

Metric	Value
Accuracy	92.01%
Precision	0.7834
Recall	0.7845
F1-Score	0.7831
Validation Accuracy	91.69%
Validation Loss	0.2784
Test Loss	0.2592

speech detection tasks when combined with practical deployment mechanisms. The final model showed clear improvement compared to early experimental versions, particularly after addressing dataset imbalance and label inconsistency issues.

One of the most important findings of this research was the impact of dataset quality on model performance. The initial model trained only on the Davidson dataset produced unreliable predictions because the dataset contained a significantly larger number of offensive samples than explicit hate speech samples. The integration of HateXplain and Jigsaw datasets significantly improved classification diversity and allowed the model to better understand different forms of harmful language.

Another important observation was that model deployment introduces challenges that are often ignored in traditional academic research. Issues such as incorrect label mapping, confidence instability, and browser integration complexity became visible only during implementation testing.

B. Comparison with Existing Systems

Most existing hate speech detection systems focus primarily on offline classification tasks. These studies often report strong accuracy metrics but do not address how such models can be deployed in real-world environments for direct user protection. Similarly, many social media platforms rely on centralized moderation systems that remove harmful content only after it has already been posted and viewed by users.

The proposed system addresses these limitations by introducing a proactive user-side protection mechanism. Instead of relying on delayed platform moderation, the browser extension detects harmful content in real time while users browse social media platforms.

Another limitation in many existing systems is the lack of transparency in prediction outputs. The proposed system improves transparency by displaying confidence scores alongside predictions, allowing users to better understand prediction reliability. Additionally, the browser extension architecture allows the proposed system to operate across multiple web platforms, making it more flexible and scalable.

C. System Strengths

One of the major strengths of the proposed system is its practical implementation. Another important strength is the use of a lightweight transformer architecture. DistilBERT provided strong contextual language understanding while maintaining

lower computational requirements compared to larger transformer models.

Another notable strength of the system is its structured API design, which supports future scalability. Since all modules within The Guardian framework communicate through standardized schemas, future integrations with additional cybersecurity modules can be implemented with minimal architectural changes.

The confidence scoring mechanism further improved system transparency by allowing users to understand prediction certainty. The risk scoring mechanism helped prioritize warnings based on content severity, which improved usability and reduced unnecessary alerts.

D. Limitations

Despite the positive results, several limitations remain. The system may struggle with sarcasm, coded language, hidden meanings, and context-dependent expressions. Hate speech often evolves rapidly, and users may intentionally modify harmful language to bypass detection systems.

The current model is also limited to English-language content. Since social media platforms contain multilingual users, this limits broader deployment capability. Another limitation involves dataset labeling inconsistencies. Even after preprocessing and standardization, some ambiguity remains between hate speech and offensive language labels.

The browser extension currently focuses only on textual content. Harmful content shared through images, videos, and memes cannot currently be detected by the system.

E. Future Improvements

Future research can improve the proposed system in several ways. One major improvement would be expanding multilingual detection capabilities to support users from diverse linguistic backgrounds. Future versions of the model could also incorporate larger and more balanced datasets to improve generalization performance.

Another important improvement involves multimodal hate speech detection. Integrating image analysis and video content analysis would allow the system to detect harmful memes and visual hate content. Advanced transformer architectures such as RoBERTa, DeBERTa, or domain-specific large language models may also improve classification accuracy. Future work could also explore adversarial text detection techniques that identify intentionally modified harmful language designed to bypass moderation systems.

F. Summary

This discussion demonstrates that the proposed research successfully addresses an important gap between academic hate speech classification research and real-world deployment. The system moves beyond traditional offline classification approaches by introducing real-time browser-based protection mechanisms that directly improve user safety.

VII. CONCLUSION AND FUTURE WORK

This research presented the design and implementation of a real-time hate speech detection framework developed to improve user safety within social media environments. The study was motivated by the increasing prevalence of harmful online interactions and the limitations of traditional moderation systems that often react only after harmful content has already reached users.

To address this limitation, this research proposed a proactive browser-based solution that detects harmful textual content in real time while users browse social media platforms. The system combines natural language processing techniques with practical cybersecurity mechanisms to reduce user exposure to harmful online communication.

The machine learning component of the system was built using a transformer-based DistilBERT model that was fine-tuned for three-class classification involving hate speech, offensive content, and neutral content. To overcome dataset limitations, additional datasets including HateXplain and Jigsaw Toxic Comment datasets were integrated into the training pipeline.

Beyond model development, a browser extension was developed to automatically detect visible textual content from social media platforms. The extension communicates with a FastAPI backend that performs real-time inference using the trained model. The final system provides visual content flagging, confidence score reporting, and risk-based user alerts.

The final evaluation results confirmed that the system successfully performs real-time content detection, harmful language classification, confidence scoring, and adaptive user warning generation. The browser extension operated with minimal delay and demonstrated the practical feasibility of deploying transformer-based hate speech detection models in real-world browsing environments.

Although the proposed system achieved its primary objectives, several opportunities remain for future improvement. Future research can explore multilingual detection capabilities, additional datasets, multimodal capabilities for images and videos, and more advanced transformer architectures. In conclusion, this research demonstrates that combining machine learning, browser technologies, and cybersecurity principles can create effective proactive defense mechanisms against harmful online communication.

REFERENCES

- [1] T. Davidson et al., "Automated Hate Speech Detection and the Problem of Offensive Language," 2017.
- [2] Z. Waseem and D. Hovy, "Hateful Symbols or Hateful People?" 2016.
- [3] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," 2019.
- [4] V. Sanh et al., "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," 2019.
- [5] B. Mathew et al., "HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection," 2021.
- [6] A. Schmidt and M. Wiegand, "A Survey on Hate Speech Detection using Natural Language Processing," 2017.
- [7] P. Fortuna and S. Nunes, "A Survey on Automatic Detection of Hate Speech in Text," 2018.
- [8] P. Badjatiya et al., "Deep Learning for Hate Speech Detection in Tweets," 2017.

- [9] Z. Zhang et al., “Detecting Hate Speech on Twitter Using a Convolution-GRU Based Deep Neural Network,” 2018.
- [10] M. Mozafari et al., “A BERT-Based Transfer Learning Approach for Hate Speech Detection in Online Social Media,” 2019.
- [11] J. Qian et al., “Hierarchical CVAE for Fine-Grained Hate Speech Classification,” 2018.
- [12] A. Founta et al., “Large Scale Crowdsourcing and Characterization of Twitter Abusive Behavior,” 2018.
- [13] B. Vidgen et al., “Measuring Hate Speech,” 2021.
- [14] T. Brown et al., “Language Models are Few-Shot Learners,” 2020.
- [15] A. Vaswani et al., “Attention Is All You Need,” 2017.